

Senior Honours Project
Effects of extracellular matrix components
on *Bacillus subtilis* biofilm's yielding
transformation

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March 2025

Abstract

The work examines the role of extracellular matrix constituents on yielding transition of *B. subtilis* biofilms. Through viscoelastic behavior comparison between wild-type and matrix constituent deletion mutants via strain-controlled amplitude sweep test and frequency sweep test, the experiment quantified elasticity variation and yielding behavior. Results reveal that matrix composition plays a significant role in biofilm stiffness and energy dissipation and is in agreement with previous prediction and experiments. However, future improvements in sample control and environment control are recommended to enhance precision.

Declaration

I declare that this project and report is my own work.

Date: 28/3/2025

Supervisor: Dr. Ryan Morris

10 Weeks

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1 Introduction

Biofilms are highly structured populations of surface-attached microorganisms. In a biofilm, microbes behave less like isolated cells and more like a cooperative community, and often exhibit specialized functions and higher tolerance to threats (e.g., antibiotics or disinfectants) than free-floating (planktonic) cells [1]. Biofilms are commonly found on wet surfaces such as ship hulls, equipment, pipelines, and medical devices, creating problems such as fouling, clogging, contamination, and corrosion. For example, microbiologically influenced corrosion due to biofilms is a critical challenge for pipelines, ships, and storage tanks and leads to high maintenance and replacement costs [2] [3]. For example, fouling by marine organisms results in boat

drag, increased fuel consumption, and frequent maintenance with the possibility of spreading invasive organisms [4]. Economically, the costs associated with biofilms are enormous in healthcare, industry, and the environment, with an estimated trillions of dollars worldwide each year [1][3]. The healthcare industry is most affected due to the chronic and antibiotic-resistant nature of biofilms that cause infections. The US National Institutes of Health (NIH) indicates that around 80% of human infections are associated with biofilms, including those related to medical implants and catheters[3]. The resulting extended hospital stays, complicated treatment protocols, and surgical procedures are major expenses within the healthcare industry, estimated at close to 100 billion per year in the United States and with around 500,000 related deaths occurring annually [1].

One of the most critical characteristics of biofilms is their rheological properties. Past work has determined that elastic and viscous modulus and yield strain play critical roles in biofilm mechanics. A high elastic modulus and a high yield strain would result in a biofilm that is stiffer and more resistant to removal since it is capable of storing more energy before it ruptures. A more viscous, compliant biofilm (lower elasticity) would be readily deformed and even be removed by flow, but would also be capable of dissipating stresses by flowing rather than by fracturing, thus avoiding sudden detachment that would remove it [5]. Furthermore, biofilms become more hydrated with increased temperature [6], becoming less rigid and more susceptible to deformation. There are still many critical aspects remain unknown despite extensive studies on biofilms. Of particular interest is the specific role of individual extracellular matrix (ECM) components in the yielding transition and the corresponding mechanisms for energy dissipation.

In order to bridge these gaps in our current knowledge, this study investigates in detail the mechanical behavior of *B. subtilis* biofilms, with specific focus on how the deletion of individual ECM components ($\Delta tasA$, $\Delta epsA - O$, $\Delta bslA$ and $\Delta pgsB$) influences the transitions. Using strain-controlled amplitude sweeps and oscillation frequency sweeps, this study identifies the yielding behavior, energy dissipation levels (G'' overshoot value) and viscoelastic properties between different mutants compared to the wild type. Through the correspondence between specific components of the ECM and specific rheological behaviors, this study expands our current knowledge about biofilm resilience and provides better means for biofilm control and removal in industrial and medical applications.

Extracellular matrix components

At the heart of the biofilm function is its extracellular matrix (ECM), a complex web that governs structural integrity and behavior of the biofilm. *B. subtilis* NCIB 3610 (wild type) is a well-established strain that naturally forms robust biofilms with rich matrix content comprising polysaccharides (*epsA - O*), amyloid protein (*tasA*), hydrophobic layer protein (*bslA*) and poly-*gamma*-glutamic acid (*PGA*) produced by the polyglutamate synthase B gene (*pgsB*). Other lab strains lose or compromise some of the characteristics of developing a biofilm and thus make 3610 WT the best representative of nature's mechanical behavior. Every matrix component plays differently in shaping the biofilm. Liu et al. (2022) [7]

investigated how *epsA* – *O*, *tasA* and *bslA* shape rheology of the biofilm. Deletion of *tasA* was found to significantly reduce viscoelasticity compared to WT while a lack of *epsA* – *O* will render a less hydrated biofilm. The biofilm is also less structured and less hydrophobic without *bslA*.

Another recent work [6] shows that PGA production also changes biofilm mechanics under some circumstances (such as higher temperature) by pulling fluid out of the substrate and creating a more liquid-like state. In addition to that, the paper also stated that *epsA* – *O* is important in limiting fluid flow through the colony. The general effects of these ECM molecules on the *B. subtilis* biofilm have been taken as a starting point to explore its transition in yield. However, it is necessary to know about the rheology that governs how these molecules respond to mechanical stress first.

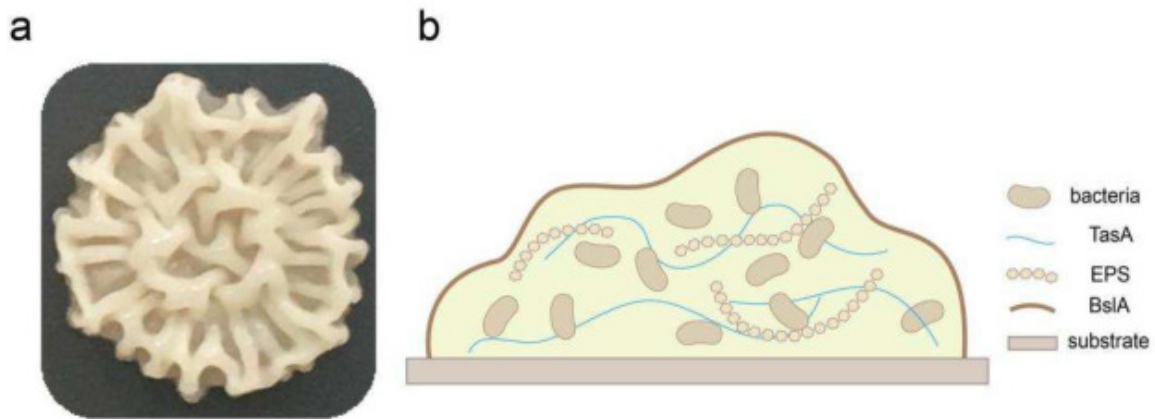


Figure 1: *Structural and compositional overview of B. subtilis biofilms [7]. (a) A 4-days-matured wrinkled colony biofilm of 3610 WT B. subtilis. (b) A schematic representation of the extracellular matrix (ECM) components that contribute to biofilm stability, including extracellular polysaccharides epsA – O, fibrillar protein tasA, hydrophobin bslA, etc.*

Viscoelastic behavior

Rheology studies flow and deformation of matter under applied external force. Depending on different rheological behaviors, materials are classified into three groups: viscous, elastic, and viscoelastic. In viscous material, external energy is completely dissipated as heat, while elastic material stores energy and deforms reversibly on application of force. A viscoelastic material exhibits viscous and elastic behavior to force, deformation, and time. Examples are polymers, gels, emulsions, and biological tissues (such as biofilms). Viscous behavior can be thought of as a dashpot based on Newton’s law, and elastic behavior can be thought of as a spring based on Hooke’s law. Hence, a viscoelastic material can be thought of as a combination of dashpots and springs. Maxwell model consists of a spring and a dashpot in series. Spring deforms instantly on constant stress application and leads to instantaneous elastic deformation. Dashpot deforms with time and leads to a permanent increase in stress. In Kelvin-Voigt model, spring and dashpot are in parallel. When stress is removed, material reverts to original form. Hence, there is no permanent deformation in this model. Hence, Maxwell model is best used to represent fluid-like behavior and

Kelvin-Voigt model to represent solid-like behavior [8]. This combination of viscous and elastic behavior forms a background for studying how the biofilm is transformed from one state to another by applied strain.

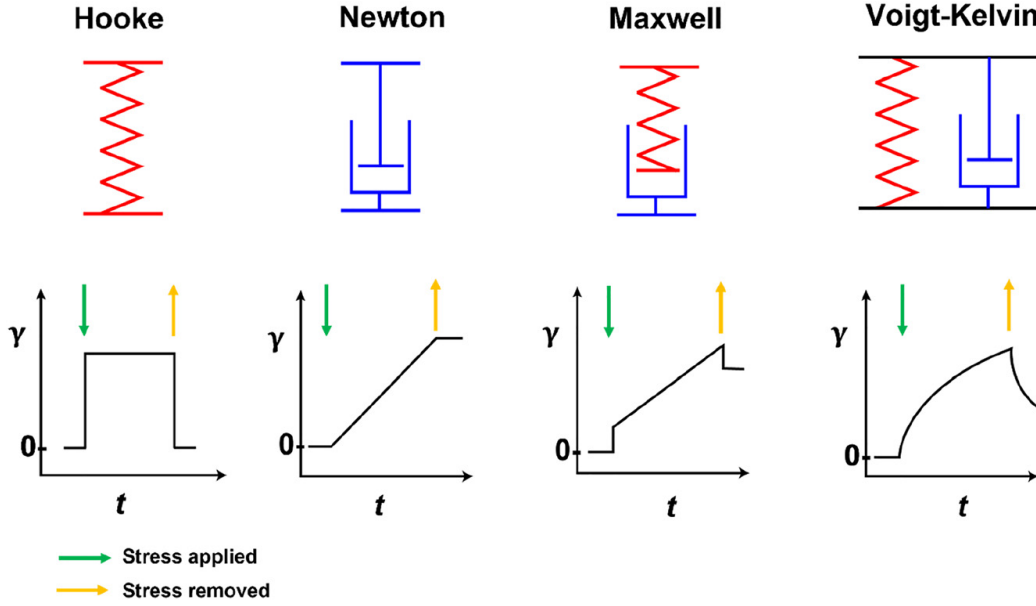


Figure 2: An illustration of Maxwell and Voigt-Kelvin's models with elastic part being red spring and viscous part being blue dashpot [8]. The jump in Maxwell model's strain indicates an instantaneous response from the spring, it is followed by a linear increase in strain due to the viscous flow of dashpot. Strain increases indefinitely as long as stress is applied in this model. In the Kelvin-Voigt model, the dashpot prevents instantaneous deformation, resulting in a gradual strain increase over time. Material under this model approaches a final strain asymptotically.

1.1 Oscillation frequency sweep

To probe the dynamic mechanical properties of the biofilm, the time-dependent behavior of the sample is measured by varying the frequency of the external force in the LVE region. Higher frequency implies a shorter period for material structural recovery and vice versa. The frequency sweep test provides valuable information on the inner structure of materials. When $G'' > G'$, the loss modulus predominates, and the material exhibits a liquid-like behavior. When $G'' < G'$, the material behaves more solid-like [8]. The crosspoint where $G' = G''$ is the gel point that represents the transition from a liquid-like state to a solid-like state.

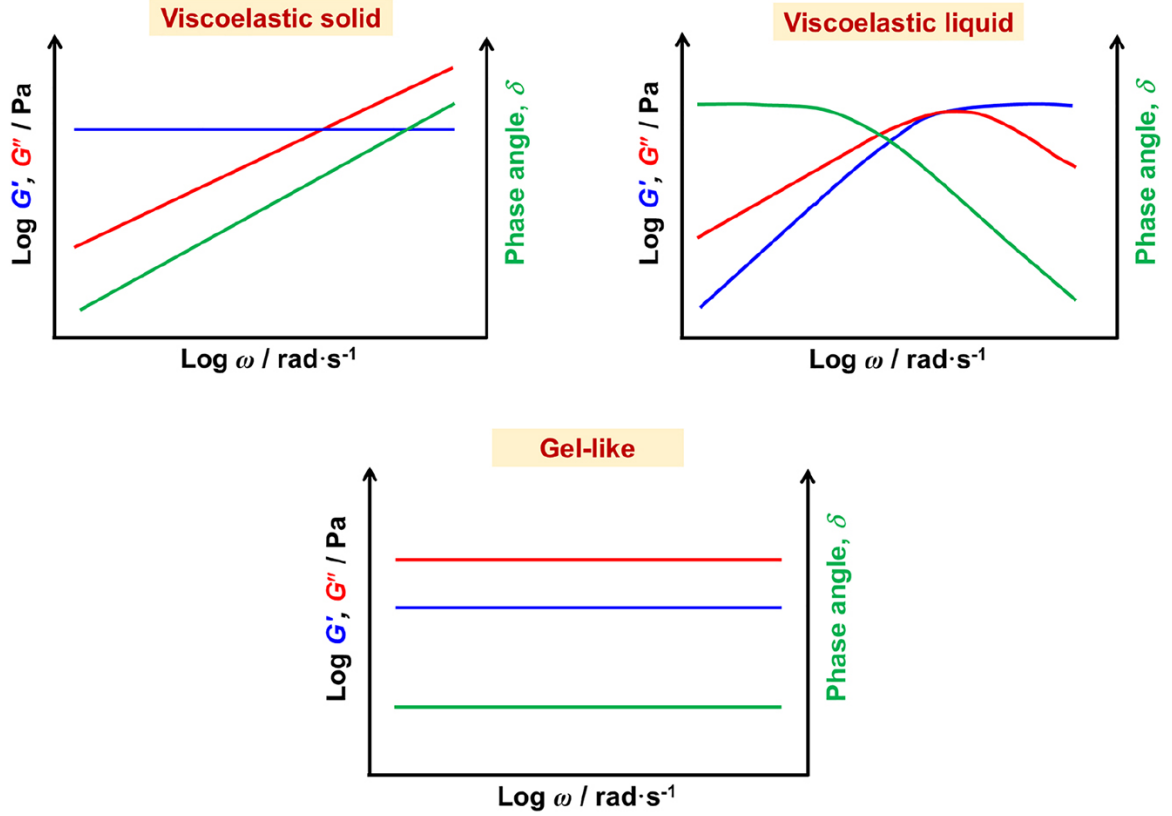


Figure 3: *Oscillation frequency sweep test plots of viscoelastic solid, viscoelastic liquid and gel-like material [8]. In a viscoelastic solid, G' is always constant and dominates at low frequency, whereas G'' dominates at higher frequency. In a viscoelastic liquid, G' is lower than G'' at low frequency but higher than G'' at high frequency. In Gel-like materials, both G' and G'' are constant through the whole frequency sweep.*

Yielding transition & Strain controlled amplitude sweep

The yielding transition from a solid state to a fluid state is governed by both viscoelastic and plastic behavior in soft materials such as gels, emulsions, colloidal suspensions, and biological tissues. Viscoelasticity can be separated into elastic response and viscous response. An elastic material under a small applied stress deforms reversibly, meaning it can return to its original shape once the stress is removed. The component of the elastic response is described by Hooke's law:

$$\sigma = G'\gamma \quad (1)$$

where σ is the stress, G' is the storage modulus and γ is the applied strain. Meanwhile, some energy is dissipated by internal friction, making the material behave like a viscous fluid. Such viscous response follows Newton's law of viscosity:

$$\sigma = \eta\dot{\gamma} \quad (2)$$

where η is the viscosity and $\dot{\gamma}$ is the strain rate. Quite a few investigations in the past employed techniques from strain-controlled oscillatory shear rheology and applied a sinusoidal signal in strain [9][10][11][12][13]:

$$\gamma(t) = \gamma_0 \sin(\omega t) \quad (3)$$

where γ_0 is the strain amplitude and ω is the angular frequency. It found out the material's response to this sinusoidal strain is given by:

$$\sigma(t) = \gamma_0 [G'(\omega) \sin(\omega t) + G''(\omega) \cos(\omega t)] \quad (4)$$

where $G'(\omega)$ is the storage modulus corresponding to the elastic component (solid-like) and $G''(\omega)$ is the loss modulus measuring the viscous response (fluid-like). The results show that the viscoelastic material behaves primarily as a solid under small strain ($G' > G''$). As the strain increases, the material dissipates more energy and eventually reaches a fluid-like state. During this process, the loss modulus shows a smooth peak and then drops rapidly as a symbol of irreversible plastic deformation. Based on the strain-controlled amplitude sweep technique, the two modulus and strain amplitude relations are obtained [13].

$$G'(\omega) = \frac{4(W_{\text{stored}}(\omega))}{\gamma_0^2} \quad (5)$$

$$G''(\omega) = \frac{2(\dot{W}_{\text{diss}}(\omega))}{\omega \gamma_0^2} \quad (6)$$

In the above equations, W_{stored} is the energy stored per unit volume and $\dot{W}_{\text{diss}}$ is the rate of energy dissipated per unit volume. For a better understanding of the physical meaning of the storage modulus G' and the loss modulus G'' , imagine a series of experiments with a bounce ball. The magnitude of G' can be visualized as the height of the bounce and the other fraction G'' is dissipated during the fall [8].

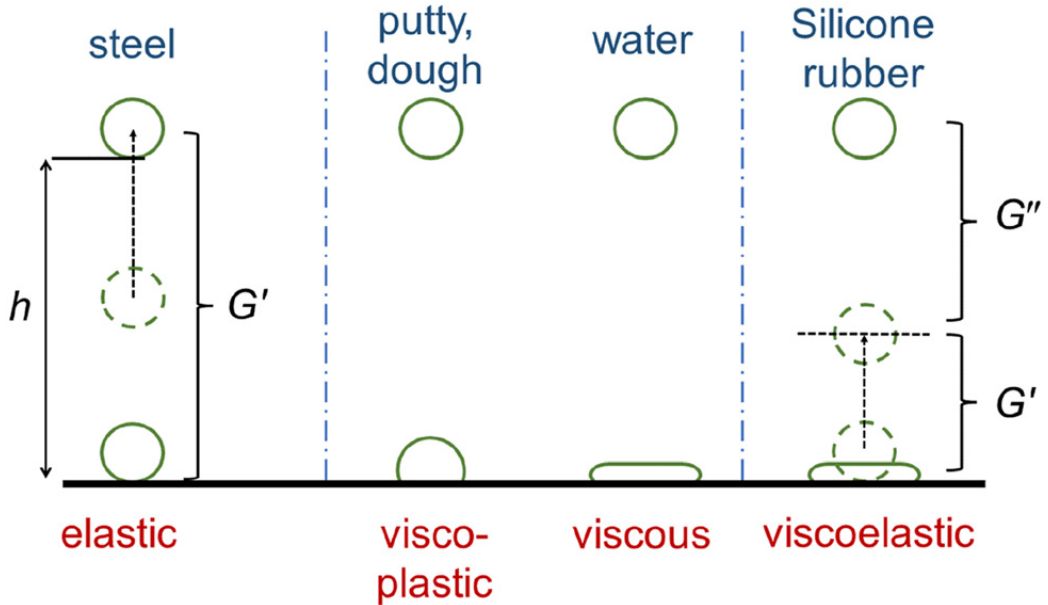


Figure 4: *Demonstration of ideal elastic response, irreversible plastic deformation, viscous flow and viscoelastic response. [8]*

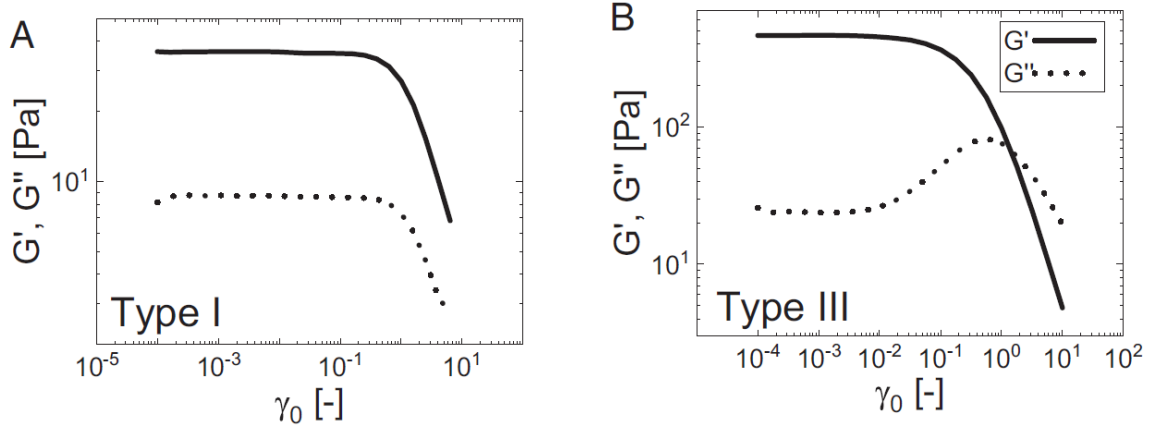


Figure 5: *Strain amplitude sweep method generated result demonstrating G'' loss modulus' behavior under increasing strain [14]. Type 1 material doesn't exhibit G'' overshoot. Materials with G'' overshoot are categorized as Type 3. Both G' and G'' show linear consistency in the low-strain region. As strain amplitude increases, G' decreases, and G'' exhibits a smooth overshoot before decreasing, indicating the yield transition.*

Linear viscoelastic (LVE) region

The insights from the transition that yield naturally lead us to define the baseline condition within the linear viscoelastic region, where the functions have constant values of G' , G'' . As demonstrated in Figure 5, such a region implies that the test carried out within it will not cause permanent deformation. The critical point of this linearity is γ_L , defined as the deviation 5% from the plateau values of G' and G'' [8].

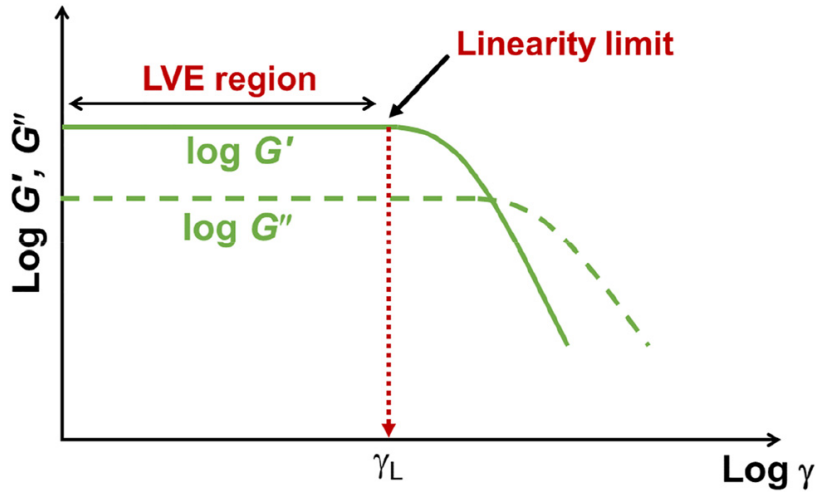


Figure 6: *Logarithmic plot of G', G'' against γ . Both G' and G'' have constant value within the LVE region, where $\gamma < \gamma_L$. [8]*

2 Methods

2.1 Biofilm preparation and growth

The foundation begins with the careful preparation and cultivation of *B. subtilis* biofilms. Luria broth (LB) consisting of 10 g of *NaCl*, 5 g yeast extract, and 10 g tryptone per liter was used for initial *B. subtilis* growth. LB was stirred at 200 RPM at 30°C for overnight growth of bacteria ($\approx 24h$). The optical density of the bacteria solution was measured the next day at the wavelength 600nm (OD600). Since LB contains various metabolites, proteins, and waste products after overnight culture, bacteria were rinsed with phosphate buffered saline (PBS) before biofilm. This removes unwanted cellular waste products to leave behind only bacteria to be used to form a biofilm. This is achieved by mixing 100 μL of cultured bacteria with 100 μL PBS and centrifuging for 3 minutes at a speed of 12 kRPM. Discarding the top waste fluid and adding 100 μL PBS to remaining bacteria was done twice to ensure that the final bacteria sample was clean.

To standardize the bacteria density, the rinsed PBS-bacteria solution must be diluted. The experiment specifies a 50% bacteria concentration. For a 1 ml sample, it requires:

$$V_{bacteria} = \frac{50\% \times 1mL}{10 \times OD} \quad (7)$$

$$V_{PBS} = 1mL - V_{bacteria} \quad (8)$$

The factor of 10 in front of the OD originates from the ratio of bacteria to PBS volume in the rinsed solution (1000 μL PBS-bacteria solution with bacteria volume: PBS $vol = 100 : 900 \mu L$). 10 μL diluted bacteria colony was evenly spaced on MSgg agar for biofilm growth. All agar used in the experiment contain 5 mM potassium phosphate and 100 mM MOPs (pH = 7.0), supplemented with 2 mM *MgCl*₂, 700 μM *CaCl*₂, 50 μM *MnCl*₂, 50 μM *FeCl*₃, 1 μM *ZnCl*₂, 2 μM thiamine, 0.5% v/v glycerol, 0.5% w/v glutamate, 1.5% w/v Select Agar (Invitrogen) [6]. Biofilm growth condition is set to 30°C for 1.5 days. With a reliable biofilm established, the next step is to configure the experimental apparatus to capture its mechanical response.

2.2 Rheology Protocol

This experiment used a Kinexus rheometer with two circular parallel plates. Plates have a rough surface to prevent wall slip, guaranteeing uniform applied shear stress. 10 to 12 colonies of biofilm were placed between the two plates. The bottom plate is stationary, while the top plate in contact with the sample under 0.1N normal force [5] rotates at constant angular frequencies or strain depending on the sweep test we performed. In the strain controlled amplitude sweep test, a sinusoidal strain was applied at a constant angular frequency of 1 Hz, while the strain amplitude γ_0 increased logarithmically from 0.01% to 200% ($\gamma_0 = 2$). This ensured that the measurement covered the LVE region and the non-linear G'' overshoot region. The storage modulus G' and the loss modulus G'' were recorded to assess the biofilm

response to increasing strain. The frequency sweep test with small constant strain amplitude $\gamma_0=0.1\%$ was applied to ensure the material remained within the LVE region. The frequency was set to decrease from 10Hz to 0.1Hz. A 2-minute rheometer calibration in air is performed between each set of experiments to ensure accuracy.

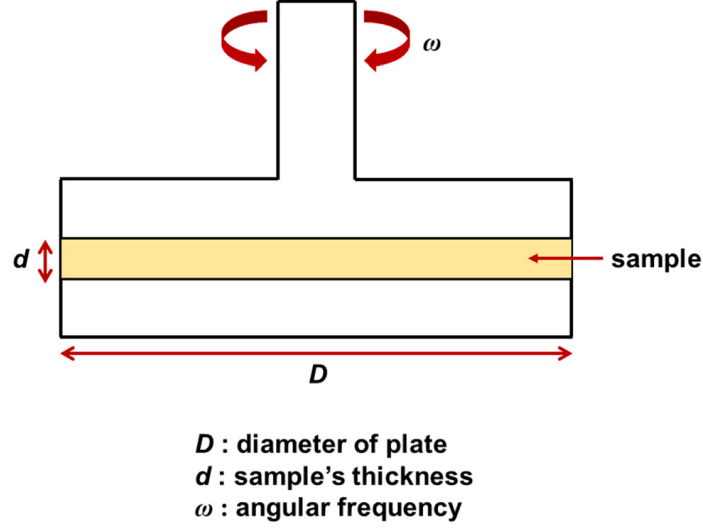


Figure 7: *A demonstration of parallel-plate geometry.*

2.3 Minimizing Edge effects

Since biofilm size varies across different types, the total amount of biofilm used in the tests is not guaranteed to be consistent. Sometimes, the biofilms could have nonuniform stress distributions. To minimize this, the rheometer will wait 60 seconds before starting the two sweep tests. During this process, the biofilm experiences a normal 0.1 N force so that it is evenly distributed between the rheometer plates.

2.4 Defining Initial G' , G'' Values & G'' Overshoot Amplitude

In order to find the linearity limit of G' and G'' in strain amplitude sweep, we need to define a reference value for each component, such that the first detected 5% deviation point from the reference value will be considered as the linearity limit. Reference values are defined as mean values of G' and G'' in the strain amplitude range [0.6%, 1%]. This is to avoid instable values in the low-strain region taking count. The G'' overshoot is defined as the difference between the global maxima of G'' and the initial G'' .

2.5 Linear interpolation

Because each replicate test produced data at different strain and frequency values, direct averaging of raw measurements of G' and G'' from different runs was not possible. To solve this, the data from each set were interpolated onto a common x-axis (e.g., the same series of strain amplitudes). Given that there are minor fluctuations

in G' , G'' throughout the strain range, we use linear interpolation rather than a more complicated cubic interpolation to avoid amplifying reading fluctuations [15]. :

$$y = y_0 + \frac{(y_2 - y_1)}{(x_2 - x_1)}(x - x_1) \quad (9)$$

where x is the data point to interpolate and y is the corresponding estimated value. Based on an existing set of data points, this method uses the equation of a line to estimate a new data point within the range of the known data points. From the above equation, we notice that $(y_2 - y_1)/(x_2 - x_1)$ is just the gradient between two known data points (x_1, y_1) and (x_2, y_2) . By interpolating all datasets onto a common x-axis using Python (see the Appendix), we ensure that all datasets contribute to the mean at the same x-values.

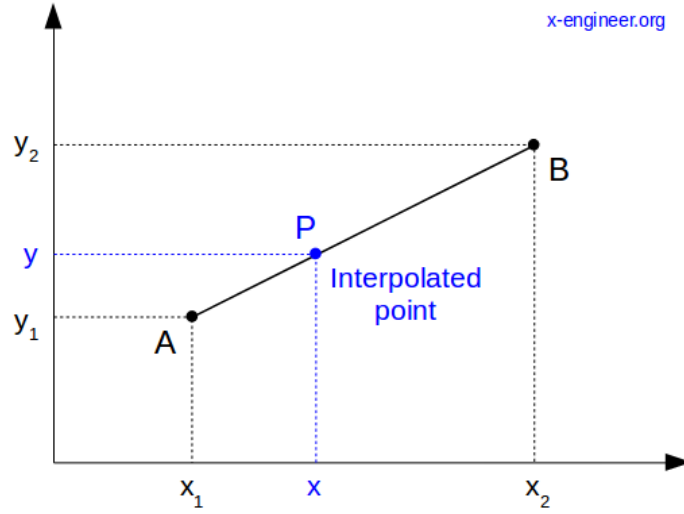


Figure 8: An illustration of linear interpolation [15]. Unknown values x and y are estimated by assuming a straight-line relationship between two known points.

2.6 Welch's t-Test

Due to the failure of the calibration of the rheometer, we have to subtract some anomalous datasets from each biofilm group, leading to different numbers of replicate files. And since there is no strong evidence that all groups share the same variance, Welch's t-test is chosen to investigate whether the magnitude and offset location of G'' overshoot has significant difference among mutant strains of *B. subtilis*.

This statistical method compares the means of two datasets and determines how likely is the difference between the means caused by random variation rather than the true effect. For a given comparison between two groups, the null hypothesis (H_0) states that there is no significant difference in the G'' overshoot between two selected groups. The alternative hypothesis states that there is a significant difference between the two groups. The t-value can be calculated as in [16]:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the means, s_1^2 and s_2^2 are the standard deviations, n_1 and n_2 are the sample sizes of two groups. In order to know our level of confidence in judging the dataset differences, we require information on the p-value for each pair of biofilm. The Welch–Satterthwaite equation computes the degrees of freedom:

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

By definition, the p-value is the probability (under the null hypothesis) of obtaining a t-statistic whose absolute value is at least as large as the observed $|t|$. Hence for a two-sided test:

$$p = 2 [1 - F_\nu(|t|)]$$

where F_ν is the cumulative distribution function (CDF) of the t-distribution with ν degrees of freedom. The final p-value is calculated using Python (see the Appendix). This experiment rejects null hypothesis if $p < 0.05$, meaning less than five times out of a hundred you would find a statistically significant difference between the means even if there was none.

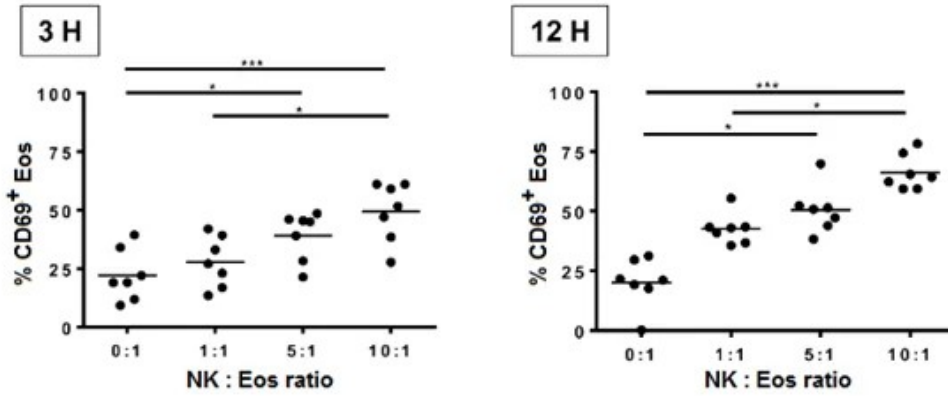


Figure 9: *Example of unpaired t-test plot: Eosinophil activation as measured by percent of CD69-positive cells after 3 and 12 hours of co-culture (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)[17]. Bars represent the standard deviations.*

3 Results & Discussion

3.1 General overview

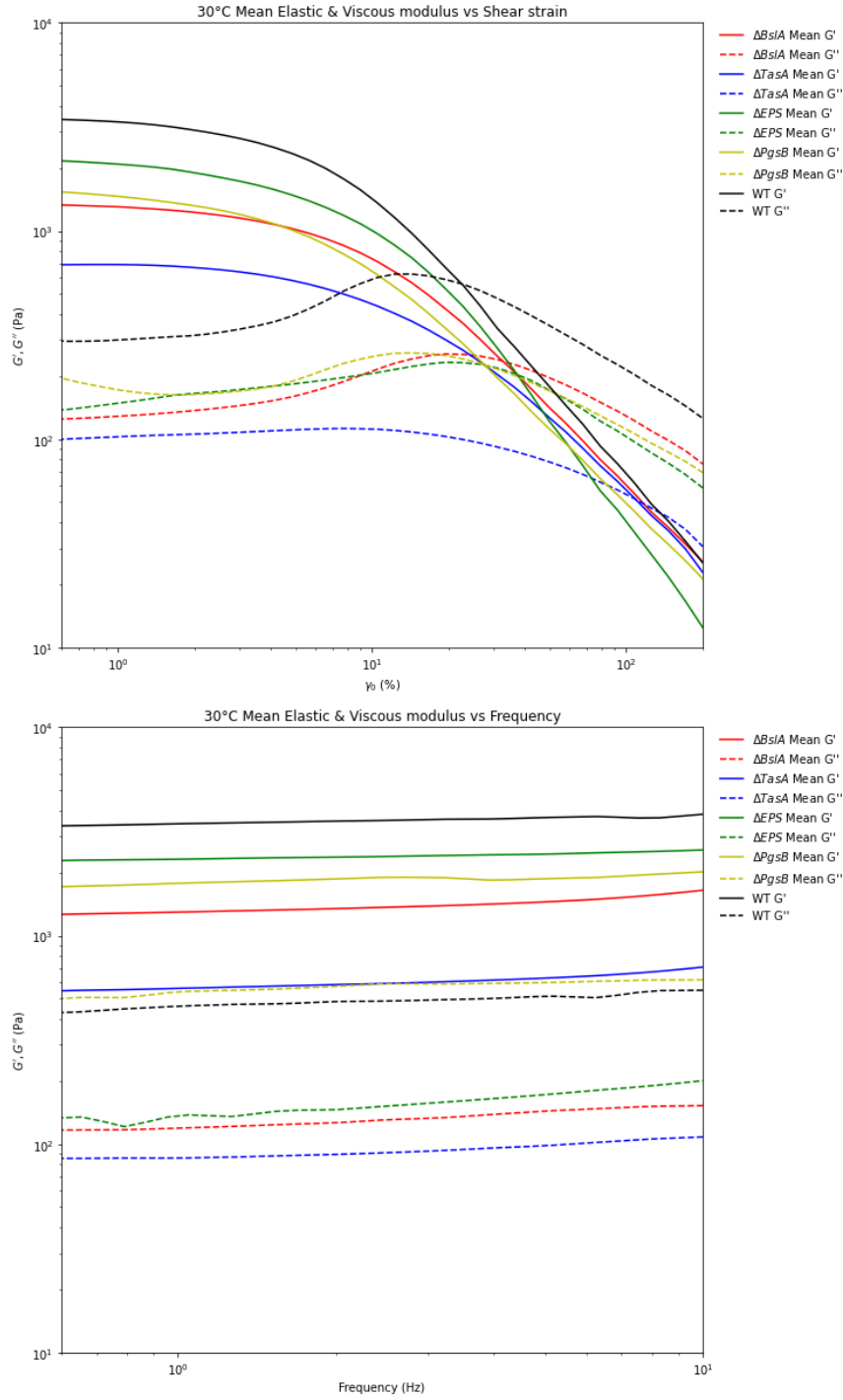


Figure 10: Two plots (Top:(a); Bottom:(b)) showing how the storage modulus G' and loss modulus G'' of the single mutant and WT vary with strain amplitude and oscillation frequency at 30°C.

Fluctuations on the readings of G' and G'' in $\gamma_0 < 0.6\%$ and $Freq. < 0.6Hz$ due to instrumental noise are ignored, so the plots are only interested in the strain range $[0.6, 200]$ (%) and the frequency range $[0.6, 10]$ (Hz).

The average values of G' and G'' of WT and other 4 single mutant strains (using the liner interpolation method) are compared for both the strain amplitude sweep and the frequency sweep as shown in Figure 10, with the solid line being G' and the dashed line being G'' . The trend of most curves of G'' in the sweep of strain amplitude follows the description of Type-3 soft material - A constant G'' LVE region followed by an overshoot and then drops down [14]. The only exception is the $\Delta tasA$ sample, whose peak value G'' is barely visible on a logarithmic scale plot. All samples' G' curves show constant values before some certain strains, and decrease once they exceed these strains (Linearity limit); this is consistent with the description of yielding transition [8]. In the frequency sweep plot (Figure 10b), we see that all biofilms have roughly constant G' and G'' , and each biofilm maintains $G' > G''$ throughout the frequency range. This follows the definition of gel-like material in Figure 3.

3.2 Initial G' & G''

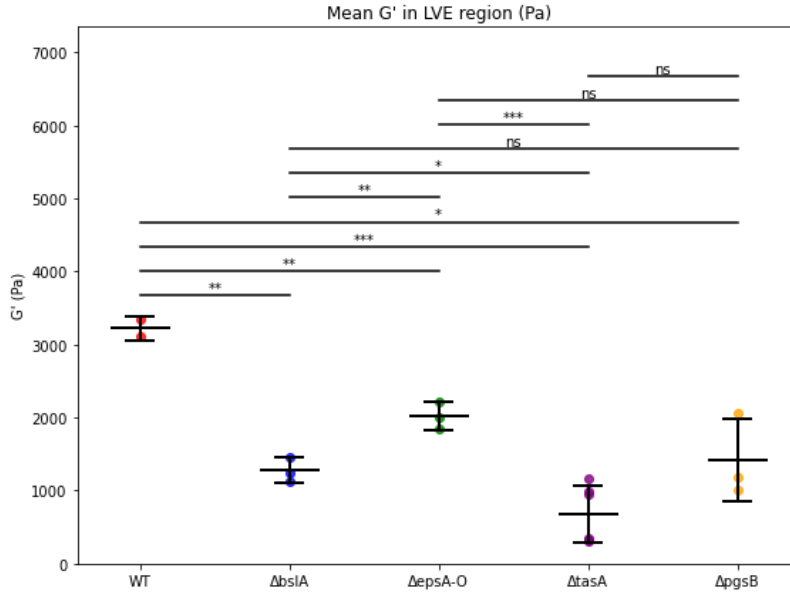


Figure 11: Mean G' in LVE strain (%) range $[0.6, 1]$. Vertical error bars represent standard deviations. Horizontal line within each error bar range represents the mean value. Dots around mean values are individual experiments' corresponding values. Asterisks indicate the t -test significance level. $p < 0.05$ (*); $p < 0.01$ (**); $p < 0.001$ (***); $p > 0.05$ (ns)

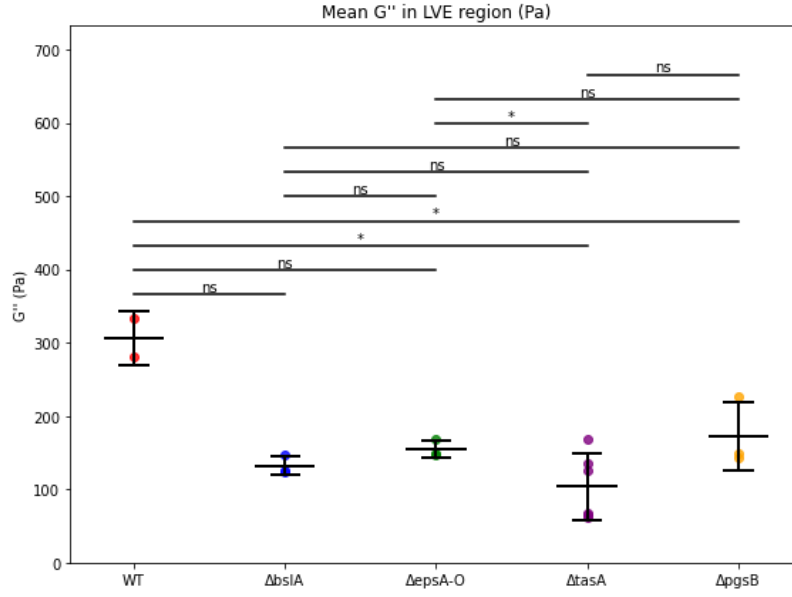


Figure 12: Mean G'' in LVE strain (%) range $[0.6, 1]$.

Figure 11 shows the initial values of G' and G'' of each biofilm in the sweep of the strain amplitude. WT has the highest initial G' ($> 3000Pa$) and the highest initial G'' ($> 300Pa$) within the LVE region, while $\Delta tasA$ exhibits the lowest initial G' ($\approx 750Pa$) and G'' ($\approx 100Pa$). Other single mutants fall into the middle.

Most p-value comparisons support that the WT has significantly higher G' and G'' compared to other mutants. However, a few comparisons (with $\Delta bslA$, Δeps) in G'' do not meet the $p = 0.05$ threshold, which means that in those cases, the WT's G'' may not be significantly different from these mutants.

3.3 Linearity Limits Comparison

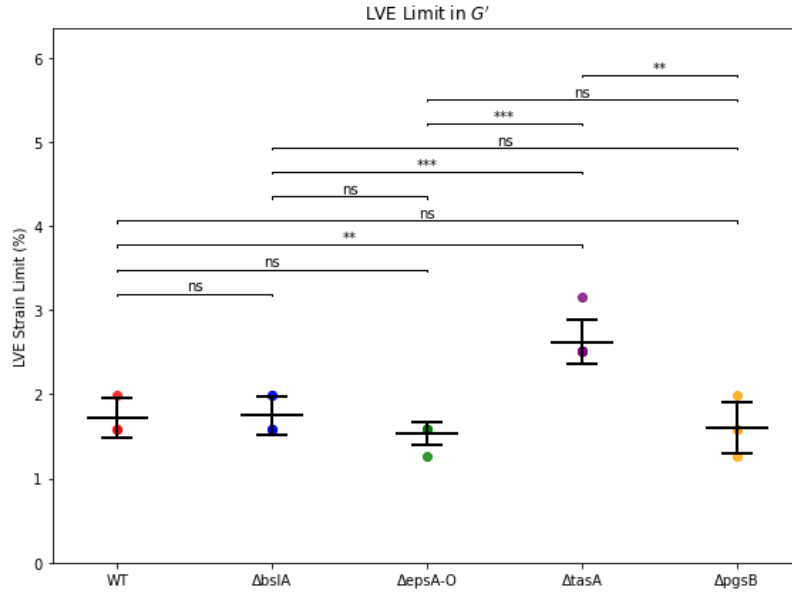


Figure 13: Mean G' LVE strain limit (%) in range $[0.6, 200]$.

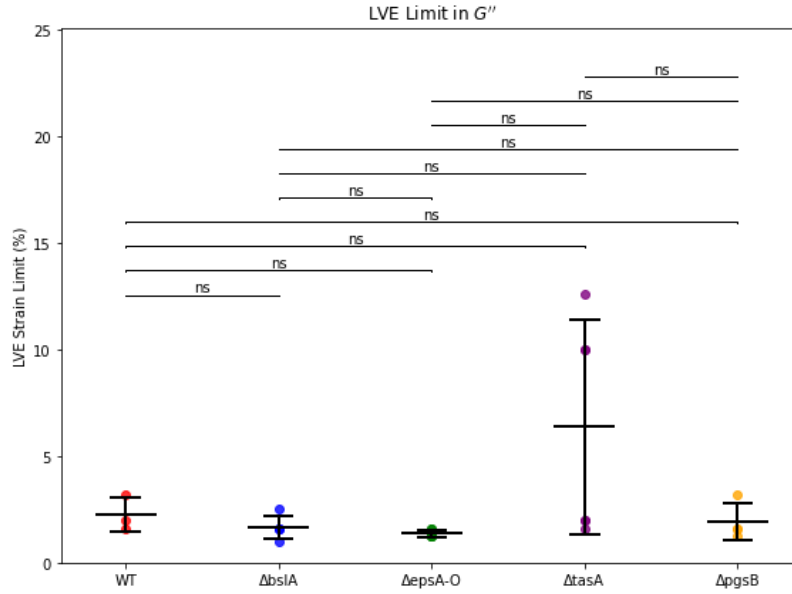


Figure 14: Mean G'' LVE strain limit (%) in range $[0.6, 200]$.

$\Delta tasA$ displays the highest average LVE strain limit in G' ($> 2.5\%$) as shown in Figure 13. This is further confirmed by the results of the p-values, where all the comparisons related to $\Delta tasA$ lie below the $p = 0.05$ threshold. For G'' , $\Delta tasA$ again shows a wider mean LVE range (LVE limit strain $> 6\%$), but with an extremely large standard deviation error bar (Figure 14). This is also reflected in the t-tests, although the p-values relating $\Delta tasA$ generally fall below 0.1, there is not enough evidence to

support this claim at the significance level $p = 0.05$ due to the large error bar in the mean limit of LVE strain $\Delta tasA$.

3.4 G'' Overshoot Magnitude Comparison

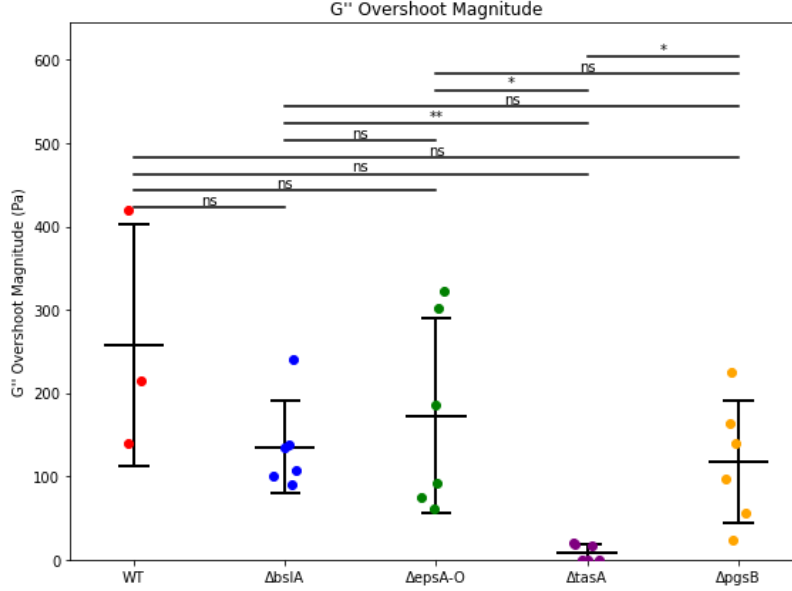


Figure 15: Mean G'' overshoot magnitude (Pa) comparison.

In Figure 15, WT shows the highest mean overshoot ($\approx 250Pa$), but the large error bar leads to considerable variability. In contrast, $\Delta tasA$ has the smallest overshoot ($\approx 10Pa$). Most p-values relating $\Delta tasA$ fall below 0.05, indicating statistically significant overshoot magnitude differences. Although the p-value of the comparison of WT and the overshoot $\Delta tasA$ is relatively low, it failed at the $p = 0.05$ threshold. This is due to the large sample variability of WT.

3.5 Transition points Comparison

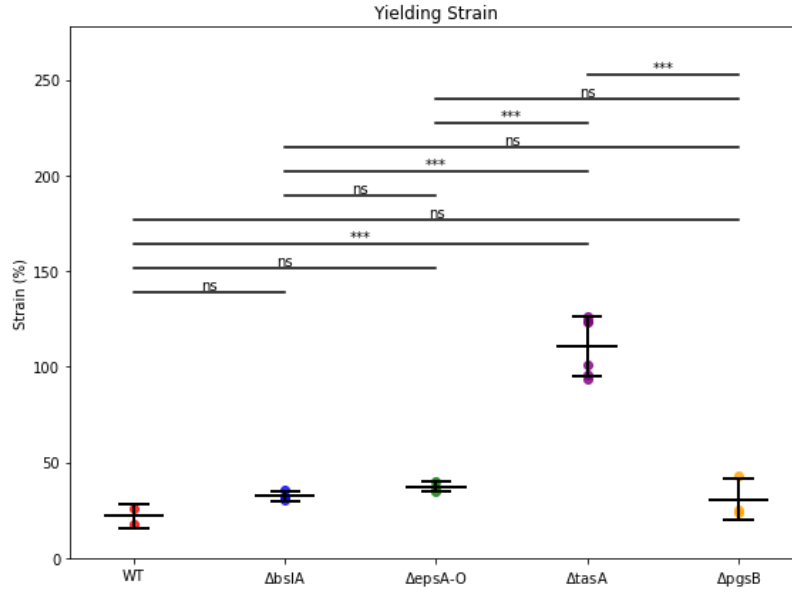


Figure 16: Mean yielding point strain (%) comparison.

Figure 16 plots the strain values at which G' and G'' curves first cross each other ($G' = G''$), representing the transition strain [14]. WT shows the lowest transition strain ($\approx 20\%$), while $\Delta tasA$ stands out with the highest transition strain ($\approx 100\%$). The p-value results have verified that $\Delta tasA$ indeed has a much higher yielding strain than other biofilms.

3.6 G''/G' comparison

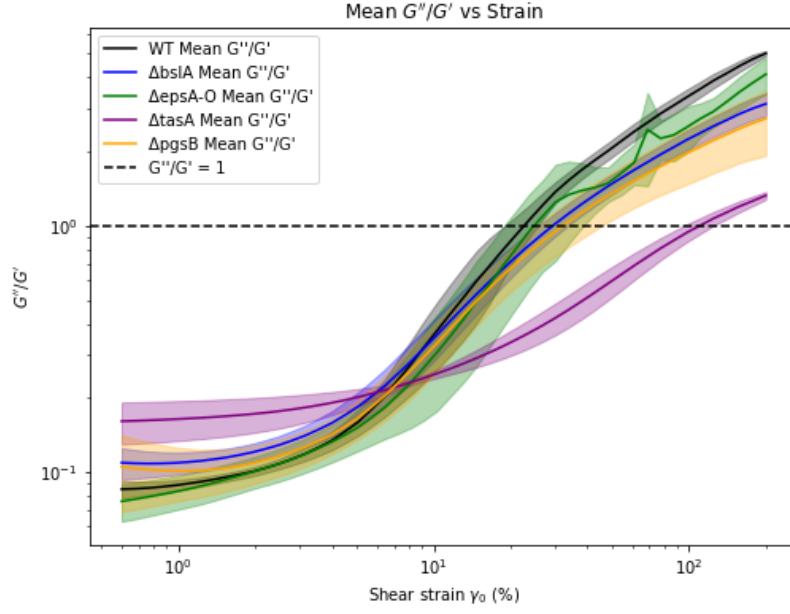


Figure 17: Linear interpolated mean G''/G' ratios in strain range $[0.6, 1]$ (%). Colour shaded region are $\pm 1SD$.

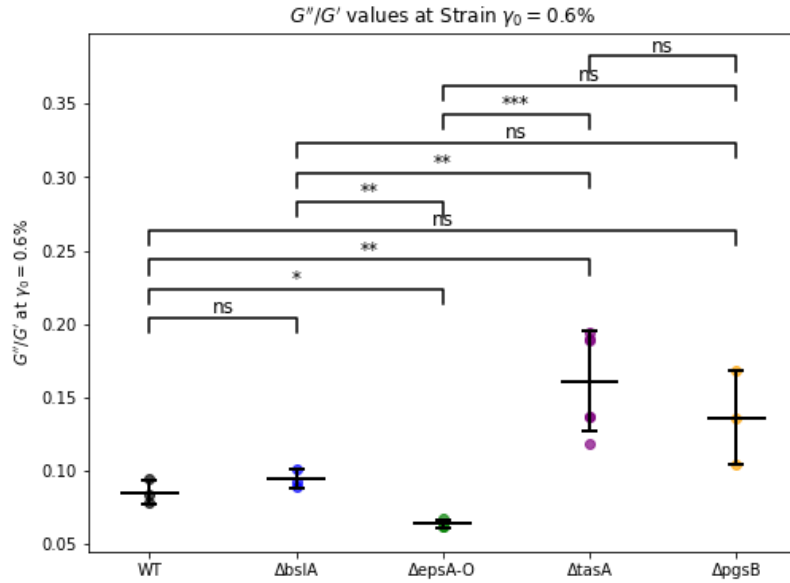


Figure 18: G''/G' ratios for all experiment runs at strain = 0.6%.

In Figure 17, all G''/G' ratios except $\Delta epsA-O$'s stay relatively constant at low strain ($< 1\%$), corresponding to the LVE region where $G' > G''$. As strain increases, all curves begin to deviate from linearity and eventually reach a point where $G''/G' = 1$ (black dashed line). In particular, $\Delta tasA$ curve has the greatest G''/G' value in the

LVE regime and the t-test suggests this high ratio stands out from other biofilms except $\Delta pgsB$.

3.7 Errors and Uncertainty

Instrumental Error

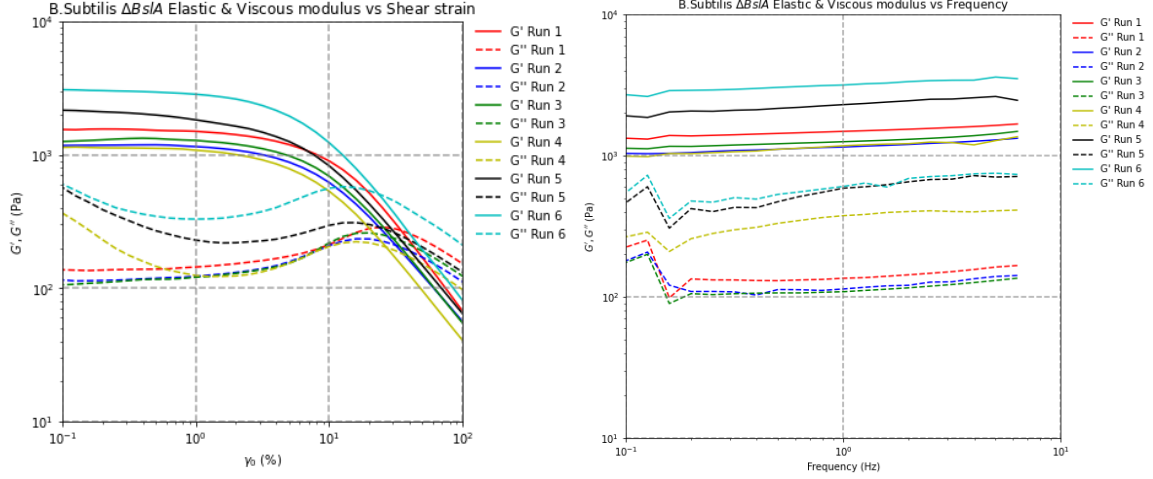


Figure 19: Left:(a) $\Delta bslA$ G' and G'' strain amplitude sweep plot, Run 1,2,3 on 31/1/25; Run 4,5,6 on 28/2/25. Right:(b) $\Delta bslA$ G' and G'' frequency sweep plot corresponds to (a).

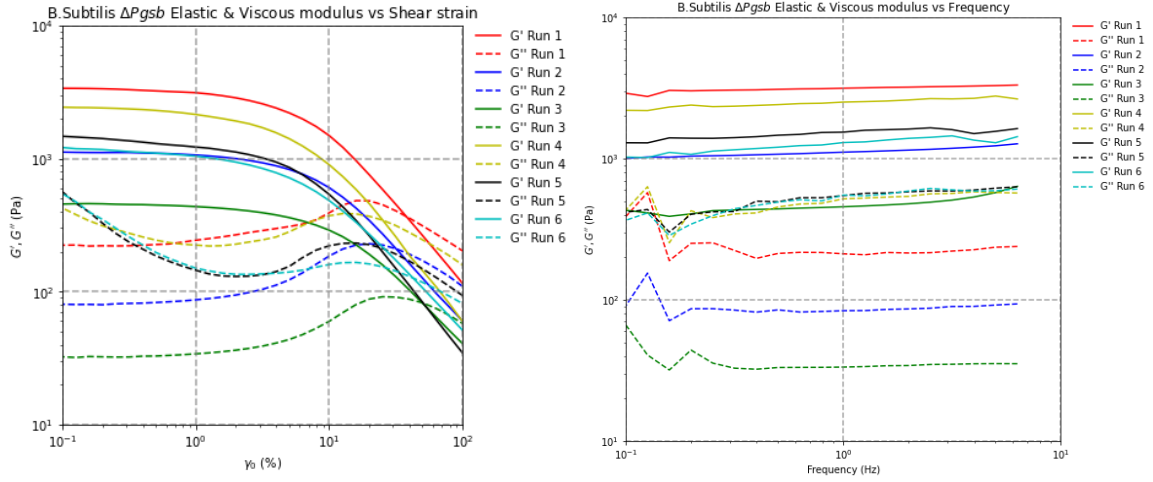


Figure 20: Left:(a) $\Delta pgsB$ G' and G'' strain amplitude sweep plot, Run 1,2,3 on 31/1/25; Run 4,5,6 on 28/2/25. Right:(b) Zoom-in photo on $\Delta pgsB$, 25/2/25.

As we have seen in Figure 11, 12, the hypothesis rejected by the t-test of $\Delta tasA$ has a significant initial difference in G' , G'' with respect to $\Delta pgsB$. In Section 3.6, we also notice that $\Delta pgsB$ has a nonlinear G''/G' behavior in low-strain regime. These can be explained by the calibration error of the rheometer. As shown in Figures 19a, 20a, in theory, G' and G'' should be constant in the low strain LVE regime[14][8],

however, run 4,5,6 of $\Delta pgsB$ and $\Delta bslA$ (performed on the same day) encountered instrumental failure during calibration, causing an anomalous decrease in G'' at low strains on the logarithmic scale plot. There is no guarantee that this would not affect our G' . G'' measurement in strain range [0.6, 1] (%) as this calibration failure exists throughout the strain range. In addition, Figures 19b, 20b show constant values G' , G'' in the frequency sweep test without any line crossing in the same run (only instrumental noise at low strains), meaning that the general trends G' and G'' in the strain amplitude plot remain valid. This calibration failure does not affect the measurement of G'' overshoot and the linearity strain limits of G' , G'' and the yielding points because the relative increase in G'' remains unchanged and the LVE limit and yielding point are strain dependent only.

Sample Variability

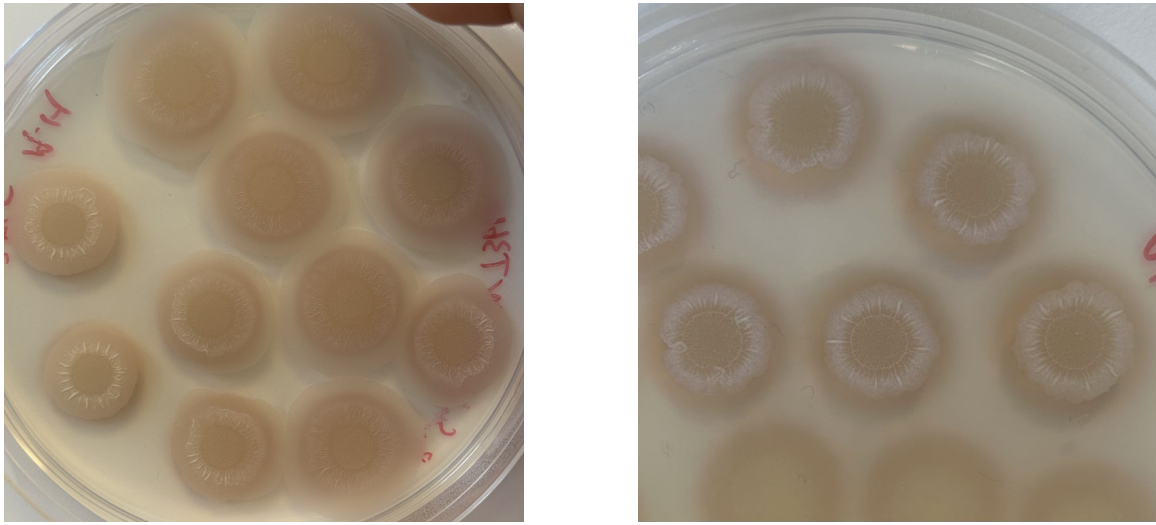


Figure 21: Left:(a) WT3610 biofilm grown on 25/2/25. Right:(b) WT3610 biofilm grown on 21/1/25.

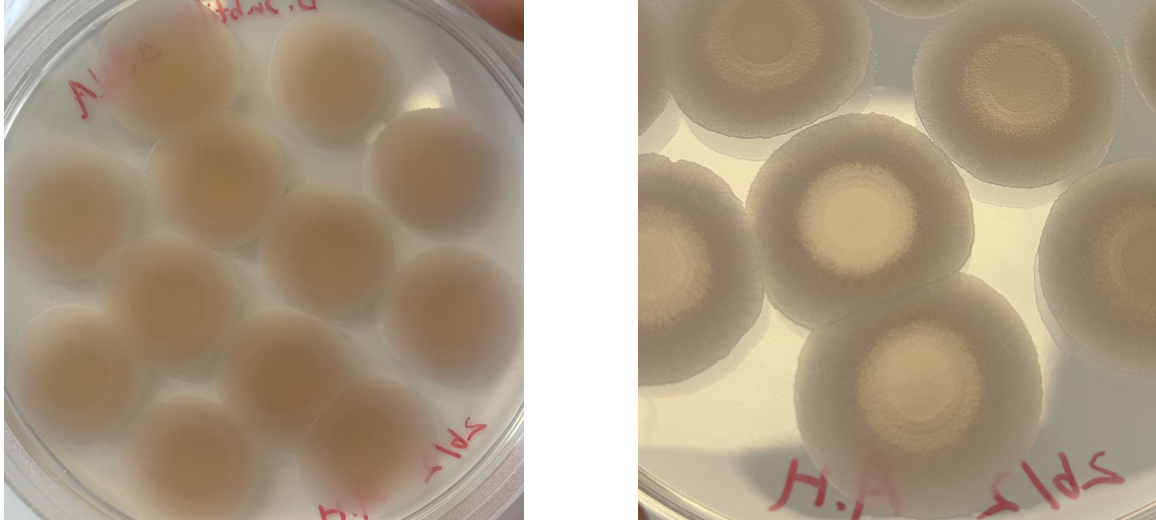


Figure 22: Left:(a) $\Delta bslA$ biofilm grown on 25/2/25. Right:(b) Zoom-in photo on $\Delta bslA$, 25/2/25.

In Figure 14, 15, we see huge standard deviations in $\Delta tasA$'s G'' LVE strain limit and the magnitude of the overshoot of WT's, $\Delta epsA-O$'s G'' . This is due to the nature of large variability in the growth and harvesting of the biological sample. Scraping the biofilm off the agar could potentially affect its native structure, leading to a softer or more compacted sample with different rheological behavior due to the external force applied during scraping. Furthermore, inconsistent biofilm thickness under 0.1 N normal force between tests can also affect the results. A thinner sample (easily compressed) could appear stiffer due to consolidation [5].

Bacteria are also sensitive to the surrounding environment. For example, as shown in Figure 21, same WT bacteria grown under the same conditions at different dates have different visual features. The bacteria colonies in Figure 21a have a large variation in size and wrinkles, while the WT colonies in Figure 21b have strong wrinkles and consistent sizes. In Figure 22, $\Delta bslA$ samples almost grew into each other. Normally, when two biofilms approach, quorum-sensing signals would trigger growth inhibition at the interface to prevent contact with other biofilms[6]. Therefore, some unknown factors are involved in the growth of bacteria (e.g., high density of bacteria drop on single agar, agar nutrient variation). These different final states of the biofilm will influence their rheological properties to some extent.

Time-dependent dehydration

In Figure 20a, we also notice an unusual curve shift along the y-axis for $\Delta pgsB$ Run 1,2,3. $\Delta pgsB$ biofilms are sensitive to humidity in the environment. Since $pgsB$ is involved in the synthesis of PGA, it is a water-attracting polymer that improves hydration[6]. $\Delta pgsB$ mutants lack PGA and struggle to absorb water from their surroundings and therefore could potentially lose water when exposed to air. A change in exposure time might have affected its rheological measurement.

3.8 Discussion

The observed LVE region is consistent with the prediction of the theory plot [14] [8] and implies that the biofilms behave as a stable elastic network with minimal energy dissipation in small strains. Within this region, no significant structural damage or rearrangements occur. The drop in G' and the temporary rise in G'' occur almost at the same strain value for most biofilms ($\approx 2\%$) except $\Delta tasA$, whose G'' LVE limit is around 3 times higher than its own G' LVE limit. It seems like the $\Delta tasA$ biofilm can deform to a greater extent while keeping the elastic behavior dominating. However, Section 3.2 shows that $\Delta tasA$ has the lowest initial G' , meaning that it is less stiff than other biofilms[8], which is consistent with Liu's prediction [7]. The unusually high LVE limit for $\Delta tasA$ is quite counterintuitive since it is less elastic than other biofilms. In the absence of $tasA$, the remaining matrix can withstand larger applied strains before the network collapses. This can indeed occur if the biofilm matrix is more loosely cross-linked. As demonstrated in Section 3.6, a higher G/G' ratio in $\Delta tasA$ indicates that the biofilm dissipates more energy relative to the energy it can store elastically. In other words, the biofilm is more fluid-like compared to others with a lower G/G' ratio. This observation has in some extent proved $\Delta tasA$ biofilms are less cross-linked. Combining with the claim that $tasA$ contributes significantly to the cohesive properties of the biofilm and helps cells to remain together [14], this phenomenon seems to make sense. But more experiments should be run to confirm the reproducibility. The high LVE limit strains in elastic and viscous modulus also result in a much delayed yielding points where $G' = G''$. As demonstrated in Section 3.5, 3.6, the completion of yielding transition for $\Delta tasA$ occurs at a significantly higher strain. Furthermore, $\Delta tasA$'s small G'' overshoot magnitude suggests that it dissipates less stored elastic energy in a more gradual way, whereas others release a burst of energy (a high G'' overshoot) when networks fail.

It is quite interesting that all $\Delta epsA - O$ results show gradually increased G''/G' before the overshoot peak occurrence (Figure 10,17) given that the rheometer was successfully calibrated. This means the $\Delta epsA - O$ biofilm partially yields at small strains rather than maintaining a stable elastic state. In other words, the yielding transition is more progressive. Typically, all biofilms should have a LVE regime, since we have already observed a gradual increase in G''/G' at $\gamma_0 = 0.6\%$, this implies that $\Delta epsA - O$ has a much narrower LVE regime below $\gamma_0 = 0.6\%$, comparing to other strains. Therefore, a more accurate (lower) strain range should be selected when analyzing the mean G', G'' within LVE region. As shown in Figure 18, $\Delta epsA - O$ has significantly low G''/G' ratio compared to other biofilms, this is consistent with theory prediction [18], which claims that $epsA - O$ play a key role in attracting and retaining water through osmotic mechanisms. The reduced hydration leads to less viscous-fluid behavior, corresponding to low G''/G' ratio.

4 Conclusion

In conclusion, our study demonstrates that *B. subtilis* biofilms exhibit distinct viscoelastic and yielding behaviors that are strongly influenced by the composition of their extracellular matrix. It is found that WT biofilm exhibit high storage (G') and

loss moduli (G'') in linear viscoelastic region, whereas $\Delta tasA$ mutants show significantly lower G' values and wider G'' LVE region and hence a delayed yielding. This suggests a less cohesive but more deformable network due to the absence of amyloid protein, which reinforces biofilm structure. Furthermore, strong evidence shows that $\Delta tasA$ biofilms exhibit no G'' overshoot in yielding, indicating biofilm behave as viscoelastic liquid with continuous energy dissipation.

On the other hand, deletions of $epsA-O$ are observed to have less effect on both G' , G'' values in LVE regime compared to other strains, but it plays a more important role during yield transition. Findings related to $\Delta bslA$ and $\Delta pgsB$ strains also provide insights in pairwise comparisons. However, more repeated experiments with better instrument calibration is suggested due to large variability in these measurements. The LVE strain limits also need to be verified so the outstanding G'' LVE strain limit in $\Delta tasA$ biofilm is verified. The universal low-strain regime applied in analysis ([0.6, 1]) needs to be revised in future for better accuracy in $epsA - O$ deletion sample measurements.

Further studies of the double-/triple-knockout experiment could potentially clarify how specific components compensate in shaping the yield transition. Moreover, extracellular DNA (eDNA) could potentially contribute to biofilm structural integrity. Through time-Resolved DNase treatment, one can apply DNase at different stages of biofilm development and measure the changes in viscoelastic parameters using oscillatory rheometry. This will tell us whether or not eDNA has a different function in initiating biofilm compared to later stages [19]. Beyond investigation of biofilm matrix mechanics, there are also areas of comparison with amyloid behavior and with neurodegenerative disease, including study of structural homologies between $tasA$ biofilm amyloid protein and disease-causing fibrils (alpha-synuclein) [20].

5 Acknowledgements

I would like to thank my supervisor, Ryan Morris, for his endless patience and enthusiastic help with all my questions.

A Appendices

```
# Linear interpolation
from scipy.interpolate import interp1d
import numpy as np

def interpolate_and_average(file_paths, x_col, y_cols, common_x, interp_method='linear'):
    all_interpolated = {y: [] for y in y_cols}

    for file_path in file_paths:
        df = pd.read_csv(file_path)

        # Remove duplicate x values to ensure unique interpolation points
        df = df.drop_duplicates(subset=[x_col])

        # If multiple y-values exist for the same x, take the mean
        df = df.groupby(x_col, as_index=False).mean()

        # Drop missing values (NaN) and sort x-axis for interpolation
        df = df.dropna(subset=[x_col] + y_cols).sort_values(by=x_col)

        # Perform interpolation for each dependent variable
        for y in y_cols:
            interp_func = interp1d(df[x_col], df[y], kind=interp_method,
                                   bounds_error=False, fill_value="extrapolate")
            all_interpolated[y].append(interp_func(common_x)) # Store interpolated values

    # Compute the mean of interpolated values across multiple datasets
    mean_values = {y: np.mean(all_interpolated[y], axis=0) for y in y_cols}

    return common_x, mean_values[y_cols[0]], mean_values[y_cols[1]]

# Welch's t-test
comparisons = list(itertools.combinations(range(len(biofilm_types)), 2))

p_values_Gpp_overshoot = []
df_Gpp_overshoot = []
labels = []

for i, j in comparisons:

    t_stat_Gpp, p_val_Gpp = stats.ttest_ind_from_stats(
        mean1=mean_Gpp_overshoot[i], std1=sd_Gpp_overshoot[i], nobs1=n_samples_Gpp_overshoot[i],
        mean2=mean_Gpp_overshoot[j], std2=sd_Gpp_overshoot[j], nobs2=n_samples_Gpp_overshoot[j],
        equal_var=False
    )
```

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